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12. Decoupling inhomogeneous systems

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More generally, we can use the decoupling method to solve an **inhomogeneous** linear system:

Given an $n \times n$ diagonalizable matrix $\mathbf{A}$:

Steps to solve $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{r}(t)$ by decoupling:

1. Find the eigenvalues of $\mathbf{A}$ (with multiplicity), and put them in a diagonal matrix $\mathbf{D}$.
2. Find a basis of each eigenspace. Since $\mathbf{A}$ is diagonalizable, there are n linearly-independent eigenvectors. Put these eigenvectors as columns of a matrix $\mathbf{S}$ in the same order of the eigenvalues in $\mathbf{D}$.
3. Substitute $\mathbf{x} = \mathbf{S}\mathbf{y}$ to get

$$\mathbf{S}\dot{\mathbf{y}} = \mathbf{A}\mathbf{S}\mathbf{y} + \mathbf{r}(t)$$

$$\mathbf{S}\dot{\mathbf{y}} = \mathbf{S}\mathbf{D}\mathbf{y} + \mathbf{r}(t)$$

$$\dot{\mathbf{y}} = \mathbf{D}\mathbf{y} + \mathbf{S}^{-1}\mathbf{r}(t).$$

(You may skip to the last equation without performing the algebra every time.) This is a decoupled system of inhomogeneous linear ODEs.

4. Solve for each coordinate function of $\mathbf{y}$.

5. Compute $\mathbf{S}\mathbf{y}$; the result is $\mathbf{x}$.

Example 12.1 Find a particular solution to $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{r}$ where $\mathbf{A} := \begin{pmatrix} -4 & -3 \\ 6 & 5 \end{pmatrix}$ and $\mathbf{r} = \begin{pmatrix} 0 \\ \cos t \end{pmatrix}$.

Solution: We will solve it instead with $\mathbf{r} = \begin{pmatrix} 0 \\ e^{it} \end{pmatrix}$ (complex replacement), and take the real part of the solution at the very end.

Step 1. We have $\text{tr}\mathbf{A} = 1$ and $\det\mathbf{A} = -20 - (-18) = -2$.

Characteristic polynomial: $\lambda^2 - \lambda - 2 = (\lambda - 2)(\lambda + 1)$.

Eigenvalues: $2, -1$. Therefore define

$$\mathbf{D} = \begin{pmatrix} 2 & 0 \\ 0 & -1 \end{pmatrix}.$$

Step 2. Calculating eigenspaces in the usual way leads to corresponding eigenvectors $\begin{pmatrix} 1 \\ -2 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \end{pmatrix}$, so define

$$\mathbf{S} = \begin{pmatrix} 1 & 1 \\ -2 & -1 \end{pmatrix}.$$

Now $\mathbf{A} = \mathbf{S}\mathbf{D}\mathbf{S}^{-1}$.

Step 3. The result of substituting $\mathbf{x} = \mathbf{S}\mathbf{y}$ is

$$\dot{\mathbf{y}} = \mathbf{D}\mathbf{y} + \mathbf{S}^{-1}\mathbf{r}.$$

We have

$$\begin{aligned} \mathbf{S}^{-1}\mathbf{r} &= \frac{1}{\det\mathbf{S}} \begin{pmatrix} -1 & -1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ e^{it} \end{pmatrix} \\ &= \begin{pmatrix} -e^{it} \\ e^{it} \end{pmatrix} \end{aligned}$$

so the decoupled system is

$$\dot{y}_1 = 2y_1 - e^{it}$$

$$\dot{y}_2 = -y_2 + e^{it}.$$

Step 4. Solving with ERF (recall the exponential response formula ERF from the course *Introduction to differential equations* lectures 9 and 10) gives the following particular solutions:

$$y_1 = \frac{-1}{i-2}e^{it} = \left(\frac{2}{5} + \frac{1}{5}i\right)e^{it}$$
$$y_2 = \frac{1}{i+1}e^{it} = \left(\frac{1}{2} - \frac{1}{2}i\right)e^{it}.$$

Step 5.

$$\mathbf{x} = \mathbf{S}\mathbf{y}$$
$$= \begin{pmatrix} 1 & 1 \\ -2 & -1 \end{pmatrix} \begin{pmatrix} \frac{2}{5} + \frac{1}{5}i \\ \frac{1}{2} - \frac{1}{2}i \end{pmatrix} e^{it}$$
$$= \begin{pmatrix} \frac{9}{10} - \frac{3}{10}i \\ -\frac{13}{10} + \frac{1}{10}i \end{pmatrix} (\cos t + i \sin t).$$

Final step. Take the real part to get a particular solution to the original system:

$$\mathbf{x} = \begin{pmatrix} \frac{9}{10}\cos t + \frac{3}{10}\sin t \\ -\frac{13}{10}\cos t - \frac{1}{10}\sin t \end{pmatrix}.$$

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